
Lecture Notes

on Exponential, Logarithms, and Half-Life

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1 Exponential Function

1.1 Definition:

An **exponential function** is a function of the form

$$y = a^x,$$

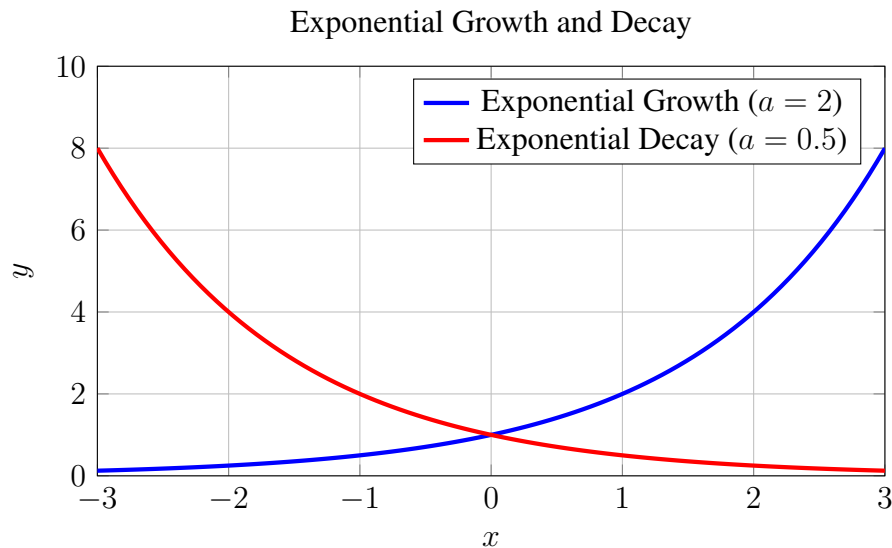
where the base a is a positive real number ($a > 0, a \neq 1$) and the exponent x is a real number.

$$a \in \mathbb{R}$$

$$x \in (-\infty, \infty)$$

1.2 Types of Exponential Functions

- **Exponential Growth:** If $a > 1$, the function $y = a^x$ increases rapidly as x increases.
- **Exponential Decay:** If $0 < a < 1$, the function $y = a^x$ decreases rapidly as x increases.



$$y = \left(\frac{1}{2}\right)^x$$

$$\text{If } a = 1, \\ y = a^x = (1)^x \\ = 1$$

$$1^0 = 1$$

Note: If $a = 1$, then the function reduces to

$$y = 1^x = 1,$$

which is simply a constant function (a horizontal line at $y = 1$), and therefore not an exponential growth or decay.

1.3 Real-World Examples of Exponential Functions

- **Finance:** Calculating compound interest over time.
- **Biology:** Studying population growth of bacteria or viruses.
- **Chemistry:** Measuring acidity levels using the pH scale.
- **Physics:** Analyzing radioactive decay of unstable elements.

1.4 Laws of Exponents *x, y are variables*

1. $a^{x+y} = a^x \cdot a^y$

2. $a^{x-y} = a^x \cdot a^{-y} = \frac{a^x}{a^y}$

3. $a^{xy} = (a^x)^y$

4. $a^x b^x = (ab)^x$

5. $a^0 = 1 \quad (a \neq 0)$

6. $a^{-x} = \frac{1}{a^x}$

For, $a=e$, $e^{x+y} = e^x \cdot e^y$

$$a^{-y} = (a^y)^{-1} = \frac{1}{a^y}$$

$$\frac{1}{2} = 2^{-1}$$

$$2^6 = 2^{2 \times 3} = (2^2)^3$$

$$3^{-2} = (3^2)^{-1} = \frac{1}{3^2}$$

Note: The general form of an exponential function is

$$y = b a^x,$$

where $b > 0$ is the initial value and $a > 0, a \neq 1$ is the base.

☛ Examples:

- Population growth: $P(t) = P_0 \cdot e^{rt}$, where $r > 0$.

r is some constant. (growth constant)

- Radioactive decay: $N(t) = N_0 \cdot e^{-\lambda t}$, where $\lambda > 0$.

λ is some constant (Decay constant)

2 Logarithm

2.1 Definition:

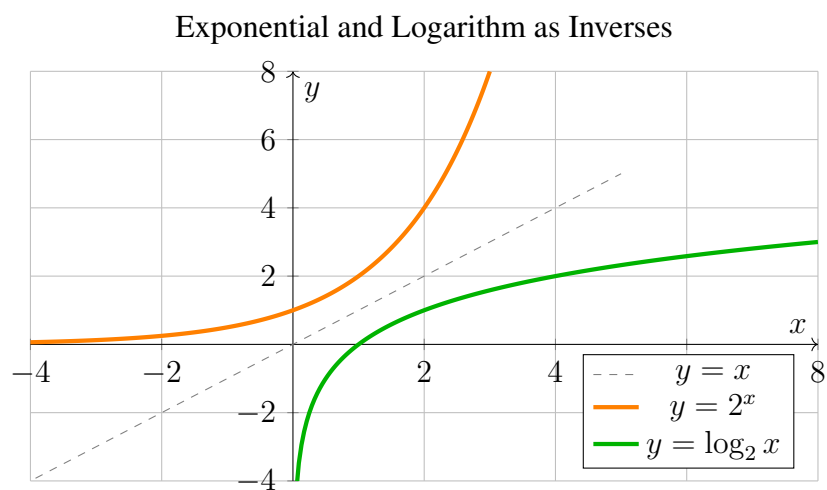
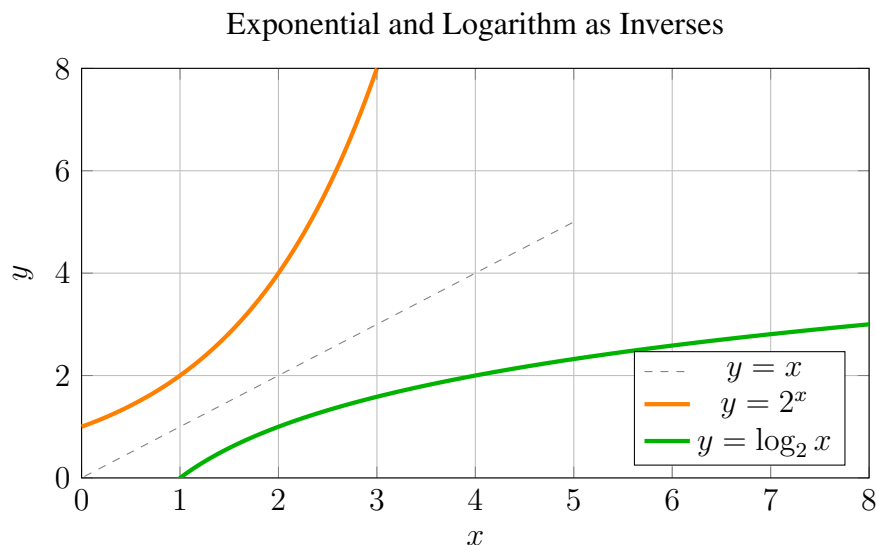
The **logarithm** of a number x to the base a (with $a > 0, a \neq 1$) is the exponent y such that

$$a^y = x.$$

It is written as

$$\log_a x = y.$$

Note: The logarithm is the **inverse** of the exponential function. That is, if $y = a^x$, then $x = \log_a y$.



2.2 Laws of Logarithms x, y are variables.

1. $\log_a(xy) = \log_a(x) + \log_a(y)$
2. $\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$
3. $\log_a(x^k) = k \log_a(x)$ (also valid if k is a variable)
4. $\log_a(a) = 1$
5. $\log_a(x) = \frac{\log_{10}(x)}{\log_{10}(a)}$ (change of base formula, base 10)
6. $\log_a(x) = \frac{\ln(x)}{\ln(a)}$ (change of base formula, base e)

$a = 10, \log(x)$

$a = e, \ln(x)$

Note: If $f(x)$ and $g(x)$ are inverse functions, then

$$f(g(x)) = g(f(x)) = x.$$

For example, let

$$f(x) = a^x, \quad g(x) = \log_a x.$$

Then, we must have (since exponential and logarithm are inverse functions)

$$f(g(x)) = g(f(x)) = x.$$

That is,

$$\text{So, } a^{\log_a x} = x$$

 **Examples:**

$$a^{\log_a x} = \log_a(a^x) = x.$$

$$\begin{aligned} f(g(x)) &= f(\log_a x) \\ &= a^{\log_a x} \\ g(f(x)) &= g(a^x) = \log_a(a^x) \end{aligned}$$

1. Solve $5^{2x} = 0.23$ for x .

$$\begin{aligned} \Rightarrow \ln(5^{2x}) &= \ln(0.23) \\ \Rightarrow 2x \ln(5) &= \ln(0.23). \quad (\text{By (2)}) \\ \Rightarrow 2x &= \frac{\ln(0.23)}{\ln(5)} \\ \Rightarrow x &= \frac{\ln(0.23)}{2 \times \ln(5)} = -0.46. \end{aligned}$$

$$\log_{10}(\quad) \quad \log_e(\quad) \quad \ln$$

2. If $\log_a x = 2.1$, and $\log_a y = 0.45$, compute $\log_a(x^3 y)$.

$$\begin{aligned} \log_a(x^3 y) &= \log_a x^3 + \log_a y \quad (\text{By 1}) \\ &= 3 \log_a x + \log_a y \quad (\text{By (3)}). \\ &= 3 \times (2.1) + 0.45 \\ &= 6.75. \end{aligned}$$

$$\begin{aligned} 3 \cdot 2.1 \times \\ 3 \times (2.1) \checkmark \end{aligned}$$

3. Solve $\log_4(3x) = 1.4$ for x .

By defⁿ: $\log_a x = y \Rightarrow x = a^y$. Recall, $a^{\log_a x} = x$

$$\begin{aligned} \hookrightarrow \log_4(3x) &= 1.4 \\ \Rightarrow 3x &= 4^{1.4} \\ \Rightarrow x &= \frac{4^{1.4}}{3} = 2.32. \end{aligned}$$

$$\begin{aligned} \therefore 4^{\log_4 3x} &= 4^{1.4} \\ \therefore 3x &= 4^{1.4} \\ \Rightarrow x &= \frac{4^{1.4}}{3} = 2.32. \end{aligned}$$

4. Simplify: $\log_2 8 + \log_2 4$.

$$\begin{aligned} &= \log_2 2^3 + \log_2 2^2 \\ &= 3 \log_2 2 + 2 \log_2 2 \\ &= (3 \times 1) + (2 \times 1) \\ &= 5. \end{aligned}$$

3 Half-Life and Doubling Time

3.1 Definition (Half-Life):

The **half-life** of a substance is the time required for its quantity to decrease to half of its initial value. If the decay is exponential, then the half-life $T_{1/2}$ is given by

$$T_{1/2} = \frac{\ln(2)}{\lambda},$$

where $\lambda > 0$ is the decay constant.

☛ **Example:** The half-life of Carbon-14 is about 5730 years, meaning that in 5730 years only half of the initial Carbon-14 atoms remain.

$$\begin{aligned} N(t=0) &= N_0 e^{-\lambda \times 0} \\ &= N_0 e^0 = N_0 \end{aligned}$$

$$\begin{aligned} \text{Let } N(t) &= N_0 e^{-\lambda t} \\ \text{at } T_{1/2}, \quad N(T_{1/2}) &= \frac{N_0}{2} \\ \Rightarrow \frac{N_0}{2} &= N_0 e^{-\lambda T_{1/2}} \Rightarrow \frac{1}{2} = e^{-\lambda T_{1/2}} \end{aligned}$$

$$T_{1/2} = \frac{\ln(2)}{\lambda}.$$

$$\begin{aligned} \Rightarrow \ln\left(\frac{1}{2}\right) &= \ln e^{-\lambda T_{1/2}} \quad (\text{By 3}) \\ \Rightarrow \ln(2^{-1}) &= -\lambda T_{1/2} \ln e \\ \Rightarrow \cancel{\ln(2)} &= \cancel{\lambda T_{1/2}} \cdot 1 \end{aligned}$$

3.2 Definition (Doubling Time):

The **doubling time** is the time required for a quantity to double its initial value under exponential growth. If the growth is exponential, then the doubling time T_d is given by

$$T_d = \frac{\ln(2)}{r},$$

where $r > 0$ is the growth rate.

✚ **Example:** If a population of bacteria doubles every 30 minutes, then its doubling time is $T_d = 30$ minutes.

✚ Example: Drug Decay in the Bloodstream

Let $C(t)$ denote the amount of a drug (in milligrams) in the bloodstream at time t (in days), and let C_0 be the initial amount of drug in the bloodstream. The decay of the drug is modeled by the function

$$C(t) = C_0 e^{-kt}, \quad (k > 0 \text{ is the decay constant}). \quad \text{--- (*)}$$

(a) If the drug has a half-life of 10 days, what is the value of k ? ≈

(b) What percent of the administered amount of drug remains in the bloodstream after 4 hours?

$$a) \quad T_{1/2} = \frac{\ln(2)}{k} \Rightarrow 10 = \frac{\ln(2)}{k} \Rightarrow k = \frac{\ln(2)}{10} \approx 0.069$$

$$\begin{aligned} (*) \Rightarrow \text{at } t = T_{1/2}, \\ \frac{C_0}{2} &= C_0 e^{-k T_{1/2}} \Rightarrow \frac{1}{2} = e^{-k T_{1/2}} \Rightarrow \ln\left(\frac{1}{2}\right) = \ln e^{-k T_{1/2}} \\ \Rightarrow \ln\left(\frac{1}{2}\right) &= -k T_{1/2} (\ln e)^{-1} \Rightarrow \ln(2^{-1}) = -k T_{1/2} \\ \Rightarrow \cancel{\ln(2)} &= \cancel{k T_{1/2}} \Rightarrow k = \frac{\ln(2)}{T_{1/2}} = \frac{\ln(2)}{10} \approx 0.069 \end{aligned}$$

$$\begin{aligned} b) \quad 24 \text{ hrs} &\rightarrow 1 \text{ day} \\ 1 \text{ hr} &\rightarrow \frac{1}{24} \text{ day} \\ 4 \text{ hrs} &\rightarrow \frac{4}{24} \text{ day} = \frac{1}{6} \text{ day}. \end{aligned}$$

$$C\left(\frac{1}{6}\right) = C_0 e^{-(0.069) \times \frac{1}{6}} \quad (\text{From } *)$$

$$\Rightarrow \frac{C\left(\frac{1}{6}\right)}{C_0} = e^{-\frac{0.069}{6}} = \exp\left(-0.069 \times \frac{1}{6}\right) = 0.988 \approx 0.99$$

$$\frac{C\left(\frac{1}{6}\right)}{C_0} \times 100 \% = (0.99 \times 100) \% = 99 \%$$

Example: Oxygen Consumption of Salmon

Biologists studying salmon have found that the oxygen (mg) consumption of yearling salmon increases exponentially with the speed of swimming, according to the function

$$f(x) = 100e^{0.6x}, \longrightarrow (**)$$

where x is the swimming speed in feet per second (ft/s).

$\Rightarrow x=0$

- (a) What is the amount of oxygen consumption when the fish are not moving?
- (b) What is the oxygen consumption at a speed of 2 ft/s?
- (c) If a salmon is swimming at 2 ft/s, how much faster does it need to swim in order to double its oxygen consumption?

a) What is $f(x)$ at $x=0$ or $f(0) = ?$

(**) $\Rightarrow f(0) = 100e^{0.6 \times 0} = 100 \text{ (mg)}.$

b) $f(2) = 100e^{0.6 \times 2} = 332.01$

c) We have to find x s.t. salmon will consume oxygen at the amount $2f(2)$.

$$\Rightarrow 2 \times f(2) = 100e^{0.6x}$$

$$\Rightarrow 2 \times (332.01) = 100e^{0.6x}$$

$$\Rightarrow \frac{2 \times 332.01}{100} = e^{0.6x} \Rightarrow 0.6x = \ln(6.64)$$

$$\Rightarrow x = \frac{\ln(6.64)}{0.6} = 3.16 \text{ ft/s}$$

$$\therefore (3.16 - 2) = 1.16 \text{ ft/s}.$$